CHAPTER 1

Numbers

INTRODUCTION

In Hindu Arabic System, we use ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called *digits* to represent any number. This is the *decimal system* where we use the numbers 0 to 9. 0 is called *insignificant digit* whereas 1, 2, 3, 4, 5, 6, 7, 8, 9 are called *significant digits*.

A group of figures, denoting a number is called a *numeral*. For a given numeral, we start from extreme right as Unit's place, Ten's place, Hundred's place and so on.

Illustration 1: We represent the number 309872546 as shown below:

Ten Crore 10^8	Crores 10^7	Ten Lakhs (million) 10^6	Lakhs 10^5	Ten Thousand 10^4	Thousand 10^3	Hundred 10^2	Ten's 10^1	Units 10^0
3	0	9	8	7	2	5	4	6

We read it as

'Thirty crores, ninety-eight lakhs, seventy-two thousands five hundred and forty-six.'

In this numeral:

The place value of 6 is $6 \times 1 = 6$

The place value of 4 is $4 \times 10 = 40$

The place value of 5 is $5 \times 100 = 500$

The place value of 2 is $2 \times 1000 = 2000$ and so on.

The face value of a digit in a number is the value itself wherever it may be.

Thus, the face value of 7 in the above numeral is 7. The face value of 6 in the above numeral is 6 and so on.

NUMBER SYSTEM

Natural Numbers

Counting numbers 1, 2, 3, 4, 5, ... are known as *natural numbers*.

The set of all natural numbers can be represented by

$$N = \{1, 2, 3, 4, 5, ...\}$$

Whole Numbers

If we include 0 among the natural numbers, then the numbers 0, 1, 2, 3, 4, 5, ... are called *whole numbers*.

The set of whole numbers can be represented by

$$W = \{0, 1, 2, 3, 4, 5, ...\}$$

Clearly, every natural number is a whole number but 0 is a whole number which is not a natural number.

Integers

All counting numbers and their negatives including zero are known as *integers*.

The set of integers can be represented by

$$Z \text{ or } I = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...\}$$

Positive Integers

The set $I^+ = \{1, 2, 3, 4, ...\}$ is the set of all *positive integers*. Clearly, positive integers and natural numbers are synonyms.

Negative Integers

The set $I^- = \{-1, -2, -3, ...\}$ is the set of all *negative integers*. 0 is neither positive nor negative.

Non-negative Integers

The set $\{0, 1, 2, 3, ...\}$ is the set of all *non-negative integers*.

Rational Numbers

The numbers of the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$, are known as *rational numbers*, e.g., $\frac{4}{7}$, $\frac{3}{2}$, $-\frac{5}{8}$, $\frac{0}{1}$, $-\frac{2}{3}$, etc.

The set of all rational numbers is denoted by Q.

That is, $Q = \{x : x = \frac{p}{q};\ p, q \in I, q \neq 0\}$

Since every natural number 'a' can be written as $\frac{a}{1}$, every natural number is a rational number. Since 0 can be written as $\frac{0}{1}$ and every non-zero integer 'a' can be written as $\frac{a}{1}$, every integer is a rational number.

Every rational number has a peculiar characteristic that when expressed in decimal form is expressible either in terminating decimals or in non-terminating repeating decimals.

For example, $\frac{1}{5} = 0.2$, $\frac{1}{3} = 0.333...$, $\frac{22}{7} = 3.1428714287$, $\frac{8}{44} = 0.181818\ ...$, etc.

The recurring decimals have been given a short notation as

$$0.333... = 0.\overline{3}$$
$$4.1555... = 4.0\overline{5}$$
$$0.323232... = 0.\overline{32}$$

Irrational Numbers

Those numbers which when expressed in decimal form are neither terminating nor repeating decimals are known as *irrational numbers*, e.g., $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi$, etc.

Note that the exact value of π is not $\frac{22}{7} \cdot \frac{22}{7}$ is rational while π is irrational number. $\frac{22}{7}$ is approximate value of π. Similarly, 3.14 is not an exact value of it.

Real Numbers

The rational and irrational numbers combined together to form *real numbers*, e.g., $\frac{13}{21}, \frac{2}{5}, -\frac{3}{7}, \sqrt{3}, 4+\sqrt{2}$, etc. are real numbers.

The set of all real numbers is denoted by R.

Note that the sum, difference or product of a rational and irrational number is irrational, e.g., $3+\sqrt{2}$, $4-\sqrt{3}$, $\frac{2}{5}-\sqrt{5}$, $4\sqrt{3}$, $-7\sqrt{5}$ are all irrational.

Even Numbers

All those numbers which are exactly divisible by 2 are called *even numbers*, e.g., 2, 6, 8, 10, etc., are even numbers.

Odd Numbers

All those numbers which are not exactly divisible by 2 are called *odd numbers*, e.g., 1, 3, 5, 7, etc., are odd numbers.

Prime Numbers

A natural number other than 1, is a *prime number* if it is divisible by 1 and itself only.

For example, each of the numbers 2, 3, 5, 7, etc., are prime numbers.

Composite Numbers

Natural numbers greater than 1 which are not prime, are known as composite numbers.

For example, each of the numbers 4, 6, 8, 9, 12, etc., are composite numbers.

Notes

1. The number 1 is neither a prime number nor a composite number.
2. 2 is the only even number which is prime.
3. Prime numbers up to 100 are:
   2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, i.e., 25 prime numbers between 1 and 100.
4. Two numbers which have only 1 as the common factor are called *co-primes* or *relatively prime* to each other, e.g., 3 and 5 are co-primes.

Note that the numbers which are relatively prime need not necessarily be prime numbers, e.g., 16 and 17 are relatively prime, although 16 is not a prime number.

BASIC FORMULAE

ADDITION AND SUBTRACTION

Addition and Subtraction is best illustrated with the help of following example:

Illustration 1: 54321 – (9876 + 8967 + 7689) = ?

Step 1 Add Ist column:

$6 + 7 + 9 = 22$

To obtain 1 at unit's place add 9 to make 31. In the answer, write 9 at unit's place and carry over 3.

$$\begin{array}{r} 54321 \\ 9876 \\ 8967 \\ 7689 \\ \hline 27789 \end{array}$$

Step 2 Add 2nd column:

$3 + 7 + 6 + 8 = 24$

To obtain 2 at ten's place, add 8 to make 32. In the answer, write 8 at ten's place and carry over 3.

Step 3 Add 3rd column:

$3 + 8 + 9 + 6 = 26$

To obtain 3 at hundred's place, add 7 to make 33. In the answer, write 7 at hundred's place and carry over 3.

Step 4 Add 4th column:

$3 + 9 + 8 + 7 = 27$

To obtain 4 at thousand's place, add 7 to make 34. In the answer, write 7 at thousand's place and carry over 3.

Step 5 5th column:

To obtain 5 at ten-thousand's place add 2 to it to make 5. In the answer, write 2 at the ten-thousand's place.

$\therefore$ 54321 – (9876 + 8967 + 7689)

= 27789.

MULTIPLICATION

01 Multiplication of a given number by 9, 99, 999, etc., that is by $10^n - 1$

Method: Put as many zeros to the right of the multiplicant as there are nines in the multiplier and from the result subtract the multiplicant and get the answer.

Illustration 2: Multiply:

(a) 3893 by 99 (b) 4327 by 999

(c) 5863 by 9999

Solution: (a) $3893 \times 99 = 389300 - 3893 = 385407$

(b) $4327 \times 999 = 4327000 - 4327 = 4322673$

(c) $5863 \times 9999 = 58630000 - 5863 = 58624137$

02 Multiplication of a given number by 11, 101, 1001, etc., that is, by $10^n + 1$.

Method: Place n zeros to the right of the multiplicant and then add the multiplicant to the number so obtained.

Illustration 3: Multiply:

(a) 4782×11 (b) 9836×101

(c) 6538×1001

Solution: (a) $4782 \times 11 = 47820 + 4782 = 52602$

(b) $9836 \times 101 = 983600 + 9836 = 993436$

(c) $6538 \times 1001 = 6538000 + 6538 = 6544538$

03 Multiplication of a given number by 15, 25, 35, etc.

Method: Double the multiplier and then multiply the multiplicant by this new number and finally divide the product by 2.

Illustrations 4: Multiply:

(a) 7054×15 (b) 3897×25

(c) 4563×35

Solution: (a) $7054 \times 15 = \frac{1}{2}(7054 \times 30)$

$= \frac{1}{2}(211620) = 105810$

(b) $3897 \times 25 = \frac{1}{2}(3897 \times 50) = \frac{1}{2}(194850)$

$= 97425$

(c) $4536 \times 35 = \frac{1}{2}(4563 \times 70) = \frac{1}{2}(319410)$

$= 159705$

04 Multiplication of a given number by 5, 25, 125, 625, etc., that is, by a number which is some power of 5.

Method: Place as many zeros to the right of the multiplicant as is the power of 5 in the multiplier, then divide the number so obtained by 2 raised to the same power as is the power of 5.

Illustration 5: Multiply:

(a) 3982×5 (b) 4739×25

(c) 7894×125 (d) 4863×625

Solution: (a) $3982 \times 2 = \dfrac{39820}{2} = 19910$

(b) $4739 \times 25 = \dfrac{473900}{2^2} = \dfrac{473900}{4} = 118475$

(c) $7894 \times 125 = \dfrac{7894000}{2^3} = \dfrac{7894000}{8}$
$= 986750$

(d) $4863 \times 625 = \dfrac{48630000}{2^4} = \dfrac{48630000}{16}$
$= 3039375$

Distributive Laws

For any three numbers a, b, c, we have

(a) $a \times b + a \times c = a \times (b + c)$

(b) $a \times b - a \times c = a \times (b - c)$

Illustration 6: $438 \times 637 + 438 \times 367 = ?$

Solution: $438 \times 637 + 438 \times 367 = 438 \times (637 + 367)$
$= 430 \times 1000$
$= 438000$

Illustration 7: $674 \times 832 - 674 \times 632 = ?$

Solution: $674 \times 832 - 674 \times 632$
$= 674 \times (832 - 632)$
$= 674 \times 200 = 134800$

SQUARES

01 To square any number ending with 5.

Method: $(A5)^2 = A(A + 1)/25$

Illustration 8:

(a) $(25)^2 = 2(2 + 1)/25 = 6/25 = 625$

(b) $(45)^2 = 4(4 + 1)/25 = 20/25 = 2025$

(c) $(85)^2 = 8(8 + 1)/25 = 72/25 = 7225$

02 To square a number in which every digit is one.

Method: Count the number of digits in the given number and start writing numbers in ascending order from one to this number and then in descending order up to one.

Illustration 9:

(a) $11^2 = 121$

(b) $111^2 = 12321$

(c) $1111^2 = 1234321$

(d) $222^2 = 2^2(111)^2 = 4(12321) = 49284$

(e) $3333^2 = 3^2(1111)^2 = 9(1234321) = 11108889$

03 To square a number which is nearer to $10x$.

Method: Use the formula:

$$x^2 = (x^2 - y^2) + y^2 = (x + y)(x - y) + y^2$$

Illustration 10:

(a) $(97)^2 = (97 + 3)(97 - 3) + 3^2$
$= 9400 + 9 = 9409$

(b) $(102)^2 = (102 - 2)(102 + 2) + 2^2$
$= 10400 + 4 = 10404$

(c) $(994)^2 = (994 + 6)(994 - 6) + 6^2$
$= 988000 + 36 = 988036$

(d) $(1005)^2 = (1005 - 5)(1005 + 5) + 5^2$
$= 1010000 + 25 = 1010025$

DIVISION

Division is repeated subtraction.

For example, when we divide 63289 by 43, it means 43 can be repeatedly subtracted 1471 times from 63289 and the remainder 36 is left.

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                1471 ← Quotient
Divisor → 43 ) 63289 ← Dividend
               43
               202
               172
                308
                301
                 79
                 43
                 36   ← Remainder
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Dividend = (Divisor × Quotient) + Remainder

or, $\text{Divisor} = \dfrac{\text{Dividend} - \text{Remainder}}{\text{Quotient}}$

Illustration 11: On dividing 7865321 by a certain number, the quotient is 33612 and the remainder is 113. Find the divisor.

Solution: $\text{Divisor} = \dfrac{\text{Dividend} - \text{Remainder}}{\text{Quotient}}$

$= \dfrac{7865321 - 113}{33612} = \dfrac{7865208}{33612} = 234$

Illustration 12: A number when divided by 315 leaves remainder 46 and the value of quotient is 7. Find the number.

Solution: Number = (Divisor × Quotient) + Remainder
$= (315 \times 7) + 46 = 2205 + 46 = 2251$

Illustration 13: Find the least number of 5 digits which is exactly divisible by 632.

Solution: The least number of 5 digits is 10000. Dividing this number by 632, the remainder is 520. So, the required number = 10000 + (632 + 520) = 10112.

$$\begin{array}{r} 15 \\ 632\overline{)\,10000} \\ \underline{632} \\ 3680 \\ \underline{3160} \\ 520 \end{array}$$

Illustration 14: Find the greatest number of 5 digits which is exactly divisible by 463.

Solution: The greatest number of 5 digits is 99999. Dividing this number by 463, the remainder is 454. So, the required number = 99999 – 454 = 99545.

$$\begin{array}{r} 215 \\ 463\overline{)\,99999} \\ \underline{926} \\ 739 \\ \underline{463} \\ 2769 \\ \underline{2315} \\ 454 \end{array}$$

Illustration 15: Find the number nearest to 13700 which is exactly divisible by 235.

Solution: On dividing the number 13700 by 235, the remainder is 70. Therefore, the nearest number to 13700, which is exactly divisible by 235 = 13700 – 70 = 13630.

$$\begin{array}{r} 58 \\ 235\overline{)\,13700} \\ \underline{1175} \\ 1950 \\ \underline{1880} \\ 70 \end{array}$$

TESTS OF DIVISIBILITY

1. **Divisibility by 2** A number is divisible by 2 if the unit's digit is zero or divisible by 2.
   For example, 4, 12, 30, 18, 102, etc., are all divisible by 2.
2. **Divisibility by 3** A number is divisible by 3 if the sum of digits in the number is divisible by 3.
   For example, the number 3792 is divisible by 3 since 3 + 7 + 9 + 2 = 21, which is divisible by 3.
3. **Divisibility by 4** A number is divisible by 4 if the number formed by the last two digits (ten's digit and unit's digit) is divisible by 4 or are both zero.
   For example, the number 2616 is divisible by 4 since 16 is divisible by 4.
4. **Divisibility by 5** A number is divisible by 5 if the unit's digit in the number is 0 or 5.
   For example, 13520, 7805, 640, 745, etc., are all divisible by 5.
5. **Divisibility by 6** A number is divisible by 6 if the number is even and sum of its digits is divisible by 3.
   For example, the number 4518 is divisible by 6 since it is even and sum of its digits 4 + 5 + 1 + 8 = 18 is divisible by 3.
6. **Divisibility by 7** The unit digit of the given number is doubled and then it is subtracted from the number obtained after omitting the unit digit. If the remainder is divisible by 7, then the given number is also divisible by 7.
   For example, consider the number 448. On doubling the unit digit 8 of 448 we get 16.
   Then, 44 – 16 = 28.
   Since 28 is divisible by 7, 448 is divisible by 7.
7. **Divisibility by 8** A number is divisible by 8, if the number formed by the last 3 digits is divisible by 8.
   For example, the number 41784 is divisible by 8 as the number formed by last three digits, i.e., 784 is divisible by 8.
8. **Divisibility by 9** A number is divisible by 9 if the sum of its digits is divisible by 9.
   For example, the number 19044 is divisible by 9 as the sum of its digits 1 + 9 + 0 + 4 + 4 = 18 is divisible by 9.
9. **Divisibility by 10** A number is divisible by 10, if it ends in zero.
   For example, the last digit of 580 is zero, therefore, 580 is divisible by 10.
10. **Divisibility by 11** A number is divisible by 11 if the difference of the sum of the digits at odd places and sum of the digits at even places is either zero or divisible by 11.
    For example, in the number 38797, the sum of the digits at odd places is 3 + 7 + 7 = 17 and the sum of the digits at even places is 8 + 9 = 17. The difference is 17 – 17 = 0, so the number is divisible by 11.
11. **Divisibility by 12** A number is divisible by 12 if it is divisible by 3 and 4.
12. **Divisibility by 18** An even number satisfying the divisibility test of 9 is divisible by 18.

13. **Divisibility by 25** A number is divisible by 25 if the number formed by the last two digits is divisible by 25 or the last two digits are zero.

For example, the number 13675 is divisible by 25 as the number formed by the last two digits is 75 which is divisible by 25.

14. **Divisibility by 88** A number is divisible by 88 if it is divisible by 11 and 8.
15. **Divisibility by 125** A number is divisible by 125 if the number formed by the last three digits is divisible by 125 or the last three digits are zero.

For example, the number 5250 is divisible by 125 as 250 is divisible by 125.

SHORT-CUT METHODS

01 **Test to find whether a given number is a prime**

Step 1 Select a least positive integer n such that n^2 > given number.

Step 2 Test the divisibility of given number by every prime number less than n.

Step 3 The given number is prime only if it is not divisible by any of these primes.

Illustration 16: Investigate, whether 571 is a prime number

Solution: Since $(23)^2 = 529 < 571$ and $(24)^2 = 576 > 571$

$\therefore \quad n = 24$

Prime numbers less than 24 are 2, 3, 5, 7, 11, 13, 17, 19, 23. Since 24 is divisible by 2, 571 is not a prime number.

Illustration 17: Investigate whether 923 is a prime number.

Solution: Since $(30)^2 = 900 < 923$ and $(31)^2 = 961 > 923$

$\therefore \quad n = 31$

Prime numbers less than 31 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Since 923 is not divisible by any of these primes, therefore, 923 is a prime number.

02 The least number, which when divided by d_1, d_2 and d_3 leaves the remainders r_1, r_2 and r_3 respectively such that $(d_1 - r_1) = (d_2 - r_2) = (d_3 - r_3)$, is = (L.C.M. of d_1, d_2 and d_3) − $(d_1 - r_1)$ or $(d_2 - r_2)$ or $(d_3 - r_3)$.

Illustration 18: Find the least number which when divided by 9, 10 and 15 leaves the remainders 4, 5 and 10, respectively.

Solution: Here, $9 - 4 = 10 - 5 = 15 - 10 = 5$

Also, L.C.M. (9, 10, 15) = 90,

$\therefore$ the required least number = $90 - 5 = 85$.

03 A number on being divided by d_1 and d_2 successively leaves the remainders r_1 and r_2, respectively. If the number is divided by $d_1 \times d_2$, then the remainder is = $(d_1 \times r_2 + r_1)$.

Illustration 19: A number on being divided by 10 and 11 successively leaves the remainders 5 and 7, respectively. Find the remainder when the same number is divided by 110.

Solution: The required remainder

$$= d_1 \times r_2 + r_1 = 10 \times 7 + 5 = 75.$$

04 To find the number of numbers divisible by a certain integer.

The method is best illustrated with the help of following example.

Illustration 20: How many numbers up to 532 are divisible by 15?

Solution: We divide 532 by 15

$$\frac{532}{15} = 35 \times 15 + 7$$

The quotient obtained is the required number of numbers. Thus, there are 35 such numbers.

Illustration 21: How many numbers up to 300 are divisible by 5 and 7 together?

Solution: L.C.M. of 5 and 7 = 35

We divide 300 by 35

$$300 = \underline{8} \times 35 + 20$$

Thus, there are 8 such numbers.

05 Two numbers when divided by a certain divisor give remainders r_1 and r_2. When their sum is divided by the same divisor, the remainder is r_3. The divisor is given by $r_1 + r_2 - r_3$.